Tablespace restoration using data pump

Export the tablespace

STEP 1: Verify the tablespace in the database

```
select file_id,tablespace_name,file_name,maxbytes/1024/1024/1024
as"MAX_SIZE",online_status,status from dba_data_files order by file_id;
```

```
 FILE_ID TABLESPACE_NAME                FILE_NAME                                                    MAX_SIZE ONLINE_ STATUS
-------- ------------------------------ ------------------------------------------------------------ ---------- ------- ---------
       1 SYSTEM                         /u01/app/oracle/oradata/PROD/system01.dbf                   31.9999847 SYSTEM  AVAILABLE
       3 SYSAUX                         /u01/app/oracle/oradata/PROD/sysaux01.dbf                   31.9999847 ONLINE  AVAILABLE
       4 UNDOTBS1                       /u01/app/oracle/oradata/PROD/undotbs01.dbf                  31.9999847 ONLINE  AVAILABLE
       7 USERS                          /u01/app/oracle/oradata/PROD/users01.dbf                    31.9999847 ONLINE  AVAILABLE
```

STEP 2: Create a new one

```
create tablespace txn_tb datafile
'/u01/app/oracle/oradata/PROD/pdb/txn_tb01.dbf' size 400m autoextend on;
```

```
    13 TXN_TB                       /u01/app/oracle/oradata/PROD/pdb/txn_tb01.dbf               31.9999847 ONLINE  AVAILABLE
```

STEP 3: Create a user under existing tablespace.

```
create user test identified by oracle default tablespace TXN_TB quota unlimited
on TXN_TB;
```

STEP4: Verify the user

```
select username,default_tablespace from dba_users where username='TEST';
```

```
SQL> select username,default_tablespace from dba_users where username='TEST';

USERNAME            DEFAULT_TABLESPACE
------------------- ------------------------------
TEST                TXN_TB
```

STEP 5: create table under this user.

```
SQL> select * from test_data;

        ID NAME
---------- ----------------------------------------------------------------------------------------------------
         1 Souvik
         2 Rahul
```

STEP 6: Now, I've taken a logical backup of the tablespace (TXN_TB)

```
expdp system/oracle@pdb directory=exp_dir dumpfile=TXN_TB_%U.dmp
tablespaces=TXN_TB
```

```
[oracle@localhost u02]$ expdp system/oracle@pdb directory=exp_dir dumpfile=TXN_TB_%U.dmp tablespaces=TXN_TB

Export: Release 19.0.0.0.0 - Production on Thu Oct 22 15:19:16 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates.  All rights reserved.

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Starting "SYSTEM"."SYS_EXPORT_TABLESPACE_01":  system/********@pdb directory=exp_dir dumpfile=TXN_TB_%U.dmp tablespaces=TXN_TB
Processing object type TABLE_EXPORT/TABLE/TABLE_DATA
Processing object type TABLE_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type TABLE_EXPORT/TABLE/STATISTICS/MARKER
Processing object type TABLE_EXPORT/TABLE/TABLE
. . exported "TEST"."TEST_DATA"                          5.546 KB       2 rows
Master table "SYSTEM"."SYS_EXPORT_TABLESPACE_01" successfully loaded/unloaded
******************************************************************************
Dump file set for SYSTEM.SYS_EXPORT_TABLESPACE_01 is:
  /u02/logical_backup/TXN_TB_01.dmp
Job "SYSTEM"."SYS_EXPORT_TABLESPACE_01" successfully completed at Thu Oct 22 15:20:41 2026 elapsed 0 00:01:18
```

STEP 7: Drop the tablespace and verify once

```
drop tablespace TXN_TB including contents and datafiles;
```

```
SQL> select file_id,tablespace_name,file_name,maxbytes/1024/1024/1024 as "MAX_SIZE",online_status,status from dba_data_files;

   FILE_ID TABLESPACE_NAME                FILE_NAME                                                  MAX_SIZE ONLINE_ STATUS
---------- ------------------------------ ---------------------------------------------------------- ---------- ------- ---------
         9 SYSTEM                         /u01/app/oracle/oradata/PROD/pdb/system01.dbf              31.9999847 SYSTEM  AVAILABLE
        10 SYSAUX                         /u01/app/oracle/oradata/PROD/pdb/sysaux01.dbf              31.9999847 ONLINE  AVAILABLE
        11 UNDOTBS1                       /u01/app/oracle/oradata/PROD/pdb/undotbs01.dbf             31.9999847 ONLINE  AVAILABLE
        12 USERS                          /u01/app/oracle/oradata/PROD/pdb/users01.dbf               31.9999847 ONLINE  AVAILABLE
```

STEP 8: Now verify the table to access or not.

```
SQL> conn
Enter user-name: test/oracle@pdb
Connected.
SQL>
SQL> select * from test_data;
select * from test_data
              *
ERROR at line 1:
ORA-00942: table or view does not exist
```

Now, import the tablespace

STEP 1: Recreate the tablespace

```
create tablespace txn_tb datafile
'/u01/app/oracle/oradata/PROD/pdb/txn_tb01.dbf'size 300m autoextend on;
```

STEP 2: Give the user quota

```
ALTER USER test QUOTA UNLIMITED ON txn_tb;
```

```
SQL> ALTER USER test QUOTA UNLIMITED ON txn_tb;

User altered.
```

STEP 3: Now, import the file

```
impdp system/oracle@pdb directory=exp_dir dumpfile=TXN_TB_01.dmp
tablespaces=txn_tb
```

```
[oracle@localhost logical_backup]$ impdp system/oracle@pdb directory=exp_dir dumpfile=TXN_TB_01.dmp tablespaces=txn_tb logfile=imp_txn_t
b.log

Import: Release 19.0.0.0.0 - Production on Thu Oct 22 15:43:30 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates.  All rights reserved.

Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Master table "SYSTEM"."SYS_IMPORT_TABLESPACE_01" successfully loaded/unloaded
Starting "SYSTEM"."SYS_IMPORT_TABLESPACE_01":  system/********@pdb directory=exp_dir dumpfile=TXN_TB_01.dmp tablespaces=txn_tb logfile=i
mp_txn_tb.log
Processing object type TABLE_EXPORT/TABLE/TABLE
Processing object type TABLE_EXPORT/TABLE/TABLE_DATA
. . imported "TEST"."TEST_DATA"                          5.546 KB       2 rows
Processing object type TABLE_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type TABLE_EXPORT/TABLE/STATISTICS/MARKER
Job "SYSTEM"."SYS_IMPORT_TABLESPACE_01" successfully completed at Thu Oct 22 15:45:27 2026 elapsed 0 00:01:50
```

STEP 4: Verify, the schema & table

```
SQL> conn
Enter user-name: test/oracle@pdb
Connected.
SQL>
SQL> select * from test_data;

        ID NAME
---------- --------------------------------------------------------------------------------------------------------------------------------
         1 Souvik
         2 Rahul
```